Capturing the Regional Signature: Calibration, Sensitivity, and Path Dependence in the NEUSv2 Atlantis Model

Joseph C Caracappa, Andrew Beet, Robert Gamble, Scott Large

2026-03-03

1 Abstract

TBD

2 Introduction

As regions strive to implement ecosystem approaches to marine resource management, the need for systematic and interdisciplinary tools for addressing present and future concerns is apparent. These approaches seek to untangle the interactions between fisheries management, environmental conditions, changes to habitat, and other stakeholder actions. End-to-end ecosystem models can improve understanding of these interconnections by simulating regional biological, environmental, and socioeconomic processes and investigating how perturbations in one aspect of the system lead to changes in another (Rose et al. (2010); Fulton (2010)). A critical step in building this capacity is developing and calibrating a model such that it replicates key processes and responds sensibly to perturbations.

Atlantis (Fulton E. A. (2004)) is an end-to-end marine ecosystem model that simulates biogeochemistry, food webs, population dynamics, and human socioeconomic systems on regional scales (Fulton et al. (2011)) . Atlantis models have been used globally to investigate strategic management questions involving human and environmental perturbations including fishing effort (Fay et al. (2019)), spatial management (Kaplan et al. (2012)), distributional shifts (Liu et al. (2025)), and climate impacts (Rovellini et al. (2025); Fay et al. (2017)). Although Atlantis models are tailored to regional needs, they generally operate on coarse spatial scales (100s to 1000s km2) but are calibrated using a wide range of available data (regional oceanographic models, fisheries-dependent data, surveys, stock assessments, and literature reviews). Recommendations of calibration criteria for Atlantis models have been suggested (Kaplan and Marshall (2016); Olsen et al. (2016)), but there is not a rigid protocol for determining whether biological processes are sufficiently realistic to address regional strategic management concerns especially given uncertainty and gaps in available data.

The Northeast United States Atlantis model was developed in 2010 Link et al. (2010)) and has been used to investigate the impacts of climate change (@Fay et al. (2017)) and alternative management actions (Link et al. (2010)) in the northeast U.S. shelf. Since then it has undergone a significant change in model structure and underlying code base leading to an updated designation NEUSv2 (Caracappa et al. (2022)). One main objective of this updated model is to inform fisheries management through model projections under alternative management scenarios. In order to improve confidence in these projections, it is necessary to calibrate the biological sub model such that NEUSv2 reproduces a suitably skillful hindcast while responding sensibly to fishery drivers. Thus, the objectives of this study were to (1) evaluate the skillfulness of the NEUSv2 hindcast with respect to predefined calibration criteria, and to quantify NEUSv2's sensitivity to (2a) changes in sustained catch, (2b) fishing-induced disturbances, and (2c) alternative catch histories.

3 Methods

3.1 Model Structure & Description

NEUSv2 a spatial model divided by 30 spatial polygons (i.e. boxes) that span the continental shelf from Cape Hatteras through the Gulf of Maine (Figure 1), and up to four vertical layers exist in each box. The model defines 89 different functional groups representing vertebrates (59), macro invertebrates (21), zooplankton (3), phytoplankton (3), and detritus (3). Of these, 47 are single species groups, encompassing all federally managed species in the region. The complete set of model parameters used can be found in the supplemental materials. A similar approach to the original NEUS Atlantis was used for biological parameterization (Link et al., 2011). For marine mammals and elasmobranchs, pupping recruitment was used, while a Beverton-Holt parameterization was used for the other vertebrate groups. All species used prescribed seasonal movement distributions based on regional habitat assessments and the NEFSC trawl survey database.

NEUSv2 physical processes (temperature, salinity, and currents) were derived from the GLORY12V1 physical reanalysis (1993-2021; Jean-Michel et al. (2021)), and size-fractionated phytoplankton biomass was forced using a regional satellite chlorophyll product (Turner et al. (2021)). Both forced variables translate to NEUSv2 accurately despite its coarse and irregular spatial structure (Caracappa et al. (2022)). Functional groups were harvested using a forced-catch time series method, where annual commercial landings were converted to a daily rate for each species. Some groups included recreational landings or discards if data was obtainable. An age-based selectivity was used such that older fish were preferred, leading to a more realistic catch-at-age distribution after a model spin-up period.

3.2 Calibration Criteria

Three main calibration criteria were used to evaluate whether individual functional groups' simulated populations were being adequately represented in NEUSv2 (Objective 1). The first, persistence, was successfully met if the functional group's domain-wide biomass never dropped below 10% of initial conditions. The persistence criteria was required for all functional groups for the hindcast to be considered calibrated, and it is commonly used as a minimum performance standard in Atlantis models (Kaplan and Marshall (2016)) as it ensures that no species go extinct within the historical period. The second criteria, stability, requires functional groups to maintain a relativized rate of biomass change under 5% yr-1. This threshold was chosen based on an analysis of NEFSC trawl survey data and stock assessment spawning stock biomass estimates, and it seeks to reduce the amount of extreme population growth in functional groups stemming from improper parameterization or disequilibrium. The third criteria, reasonability, is met if functional group biomass in the last 20 years of the simulation (2001-2021) lies within the historical range of catchability-corrected population biomass estimates from the NEFSC trawl survey database. The reasonability criteria ensures that the mean state of the population is within the observed range. Due to uncertainty inherent in reference data sources, the stability and reasonability criteria were not necessarily required of all species for the NEUSv2 hindcast to be considered calibrated.

3.3 Sustained Fishing Experiment

To address Objective 2a, simulation experiments were performed that evaluated how sensitive each fished functional group in NEUSv2 was to sustained changes in fishing pressure. This was performed by running the full hindcast simulation (1998-2021) then projecting for 20 years at different multiplicative scalars of contemporary catch. The mean of the last 10 hindcast years (2011-2021) were used as the baseline catch. The scalars, S = 2, 5, 10, 50, and 100 were chosen to ensure that each group would be entirely depleted for at least one level. Sustained fishing experiments were performed for each single-species functional group that was exploited in NEUSv2 (N = 40) for each catch scalar level (M = 5) resulting in 200 simulations. Only single-species functional groups were chosen due to incomplete information on multi-species functional groups and no calibration expectations on depletion rates or response to fishing pressure.

Two sensitivity metrics were used to characterize each species' response to sustained changes in fishing pressure. The first was a measure of additional scope for exploitation and defined by the maximum scalar of contemporary catch that allowed the persistence criteria to be met. The second was a measure of responsiveness defined by the steepness coefficient R. R is estimated by fitting the following functional form using non-linear least-squares regression:

$$A_{rel} \text{ or } B_{rel} = (a + RS^c)^{-1} + d \tag{1}$$

where B_{rel} or A_{rel} are the biomass (or abundance) relative to the unperturbed scenario, a^{-1} is the upper bound, c is a concavity coefficient, d is the lower asymptote, and R is a steepness coefficient. This form was a better representation of the depletion behavior seen in experiment results and resulted in a lower AIC than a simple exponential decay function (see supplement). To ensure that R described similar sensitivities for each response variable, all species' $R_{biomass}$ and $R_{abundance}$ were compared with a linear regression. It was hypothesized that more exploited species would be more sensitive to changes in fishing pressure, so a linear regression between $R_{biomass}$ (and $R_{abundance}$) and contemporary exploitation F_0 was performed, where $F_0 = \bar{B}_0/\bar{C}_0$ and B_0 and C_0 are the biomass and catch over the last 5 hindcast years.

3.4 Disturbance-Recovery Experiment

To address Objective 2b, a second set of simulation experiments were performed that explored each functional groups' response to fishing-induced disturbances and the following recovery period. This was performed by simulating the full hindcast period (1998-2021), followed by a 2 year shift in catch (t_0 to t_1), then a 15 year recovery period (t_15). For each species, the same catch scalars on contemporary catch were used (S = 2, 5, 10, 50, 100) as in the sustained fishing experiment (Figure 5), but after the disturbance ended, fishing was resumed at contemporary catch levels (Fig. XXX). This experiment was performed for all exploited single-species functional groups (N=5) for each disturbance scalar (M=5), resulting in 200 simulations.

For all disturbance-recovery scenarios, the impact (I) at time t and disturbance magnitude (s) was calculated as

$$I_{s,t} = \frac{B_{s,t}}{B'_t} \tag{2}$$

where $B_{s,t}$ is the scenario biomass, and B'_t is the biomass in the unperturbed scenario. The disturbance impact was calculated at the end of the 2 year disturbance $I_{s,t1}$ as well as 15 years post-disturbance $I_{s,t15}$. Then the proportional recovery $P_{s,t15}$ normalized to initial disturbance was calculated at t_{15} as

$$P_{s,t15} = \frac{I_{s,t15} - I_{s,t1}}{I_{s,t1}} \tag{3}$$

3.5 Path Dependence Experiment

Objective 2c addresses a possibility that similar NEUSv2 hindcast results could be obtained with any catch forcing with a similar mean and variance. However, if NEUSv2 were highly sensitive to the order of historical events, it would improve confidence that the NEUSv2 hindcast was capturing a specific regional signature (i.e. path dependence). Alternative catch forcing scenarios were constructed to determine whether the final model state was sensitive to the ordering of historical events. These scenarios simulated the model unaltered for the first 20 years, then generating a new catch forcing by permuting the annual catch forcing for the remaining years, followed by a 5 year projection at contemporary catch levels. This allowed each NEUSv2 scenario to experience an alternative ordering of historical fishing with a common 5-year period over which to compare. A total of 750 permutation scenarios were performed. Then the mean biomass ($B_{i,,s}$) for each species (s) was calculated over the last 5 years for each permutation scenario (i). Then the 95% credible interval was calculated for all $B_{i,s}$. If the historical simulation biomass for that species (B'_s) lies outside that interval, we considered species s path dependent. To quantify the degree of path dependence the absolute deviation relative to the median of B_i,s was calculated (D_s). Species with larger D_s were path dependent (i.e. more sensitive to historical fishing patterns). All statistical analyses were performed in R.

All statistical analysis were performed in R.

4 Results

4.1 Model Calibration Status

Of all functional groups in NEUSv2, 100% met persistence criteria, 84% met stability criteria, and 72% met reasonability criteria (Table 1). A summary of biomass, fishing mortality, and age structure can be found in Figure S1. Most groups show overall positive population growth rates, and of the groups that failed stability criteria, the mean relative growth rate was 21% yr-1. While not a quantitative calibration criteria, it was desired to have a stable long-term age-structure in all vertebrate functional groups. Most species maintained a relatively flat juvenile-to-adult ratio throughout the hindcast, with the exception of periods with high fishing mortality. Of the groups that failed reasonability criteria, most were multi-species functional groups with non-comprehensive reference data. Of the single-species functional groups that fail reasonability, all have population biomasses above the upper observation bound, and include Atlantic mackerel (x1.4), black sea bass (x1.2), Atlantic salmon (x59.8), and Atlantic halibut (x1.26).

4.2 Sustained Fishing Experiment

For all species, the relative biomass (N = 57) or abundance (N=45) as a function of fishing scalar was modeled by asymptotic decay (Eq.1) or a linear regression. Most species (75% by biomass; 80 by abundance) were better represented by the asymptotic decal model based on AIC. The species that were better modeled by linear regressions did not experience sufficient catch to cause depletion. Sensitivity metrics ($R_{biomass}$ and $R_{abundance}$) were in agreement, with a slope of 1.00 ± 0.03 (p « 0.01; Figure 2). The species SSH was removed from this analysis as it was a major outlier.

It was hypothesized that more exploited species (higher F_0) in the baseline simulation would experience higher sensitivity to changes in fishing pressure. Since F0 is a measure of the realized exploitation rate for simulated populations, it is not necessarily equal to true fishing mortality. A linear regression showed that Rbiomass (Figure 3a) and Rabundance (Figure 3b) were both positively correlated with F0 (biomass: $R^2 = 0.87$, p « 0.01; abundance: $R^2 = 0.67$, p «0.01). Generally, apex predators and planktivores exhibited low F_0 and low R, while piscivore and benthivore species were more variable.

4.3 Disturbance-Recovery Experiment

An initial disturbance magnitude (I_{t1}) was seen after a 2 year disturbance (Figure 4) in all species except for both squid groups (Figure 5). Most other groups showed detectible initial disturbances with I_{t1} significantly less than 1 at all disturbances sizes. Those that were generally insensitive to disturbances had low F_0 in baseline simulations such that increased scalars of catch resulted in minor overall removals. Two species (RED and TAU) exhibited $I_{t1} > 1$, indicating positive population growth after recovery. BLF, HAD, and BSB were the most sensitive to disturbances, where S = 5 scenarios completely depleting each population.

After the 15 year recovery period (Figure 7), most species show positive recoveries S < 10. For larger S, most species do not show positive recovery or are extinct. This general pattern indicates a window of disturbance size where species can recover, disturbances up to a limit were largely unrecoverable. This pattern is not seen in species that were initially insensitive to disturbances. Some benthic invertebrates (LOB and SCA) decline at all disturbance scalars.

4.4 Path Dependence Experiment

To evaluate a path-dependent signature, the baseline biomass (B') of each species was compared to the credible interval of the biomass distribution generated by resampled catch inputs (B_i). Of fished-species, 76% of showed path-dependency, with 80% of all species (Table 2). All apex-predators, 81% of benthivores, and 50% of piscivores and planktivores were path-dependent with regards to catch history (Figure 7). All guilds showed significant deviation from the mean of their respective B_i, with most species 1 to 10 standard deviations from the mean. The general pattern of strong path-dependence in NEUSv2 species indicates that simulated contemporary conditions, regardless of model skill, are a unique product of the regional catch history.

5 Discussion

5.1 Model Calibration

The simulated functional groups in NEUSv2 generally meet the defined calibration criteria. However, since some species are not well represented by the NEFSC trawl survey, the biomass reference points used in the reasonability criteria possess varying degrees of uncertainty. NEUSv2 was also calibrated using a simple forced-catch fishery, where catch is removed from the populations in equal proportions to their spatial distribution. In reality, fishing footprints do not always mimic the population distribution. Catch is also applied as a daily rate, so seasonal patterns in fishing effort are not captured. Further development of NEUSv2 seeks to improve these fisheries processes by simulating more dynamic fleet behavior and specificity.

5.2 Fishing Sensitivity

Most species respond nonlinearly to increases in exploitation in NEUSv2. For these species, biomass begins to decline linearly with increases in catch, but since the selectivity of very young fish is low, biomass stabilizes at a low value even at high catch scalars. Generally, the realized exploitation rates that populations can sustain in NEUSv2 is much higher than what they experience in the real system, likely due to a higher simulated biomass. However, the positive correlation between exploitation and population biomass is expected. The novel sensitivity metric defined here performs similarly whether using biomass or abundance, with few exceptions. Species with linear declines in catch possess high simulated biomass and low catch, resulting in their populations not experiencing depletion even at high catch scalars. We hypothesized that more exploited stocks would be more sensitive to increases in fishing pressure, and this was confirmed by this sensitivity metric. In the future, it may be useful to revisit such metrics over catch perturbations more relevant to fisheries management measures, especially as fishing behavior is refined.

In the disturbance-recovery experiment, nearly all NEUSv2 functional groups are able to recover from temporary increases in catch. This behavior is indicative of inherent stability and elasticity in the simulated species, such that moderate perturbations in species biomass do not result in systemic collapse. While a moderately stable model is necessary to test alternative environmental and human impacts, an ecosystem model that is too stable may not be able to reproduce alternative stable states (e.g. regime shifts). Most species in NEUSv2 respond with mild to moderate catch perturbations by converging to the unperturbed baseline simulation within twenty years. However, some species show signs of a tipping point, whereby disturbances over some threshold are unrecoverable for the species. We also observe some species recovering and stabilizing at a different population biomass than the baseline simulation. These results suggest NEUSv2 might be used to explore regime shifts and tipping point dynamics in the future.

5.3 Path Dependency

The path dependency method used in this experiment is a novel application for an Atlantis model and serves as a useful test for ensuring that the model is calibrated specifically to historical events (e.g. fisheries management decision, environmental changes, market drivers, etc.) and is not merely representing some mean system state. Although, it is not possible to attribute simulation output to specific historical fisheries management decisions given the dynamic, complex, and simultaneous processes occurring within an end-to-end model simulation. This path dependency test does not require this specific attribution, but more generally indicates that the ordering of historical events contained within the forcing time series is imparting some uniqueness onto the model output as a whole. Given the recent advancements in parallel computing capacity, such path dependency tests should be repeated for other forcing variables (e.g. primary production, temperature, etc.) to ensure that Atlantis model hindcasts are a unique response to historical changes in physical environments.

6 Conclusion

The Northeast U.S. Atlantis ecosystem model (NEUSv2) was designed to improve understanding of ecological and socioeconomic interactions as well as create projections under alternative management actions and climate change scenarios. Given this broad focus, it is important that NEUSv2 preserves key ecological and fisheries processes while also being adequately calibrated. However, model assumptions and implications inherent in the Atlantis modeling framework and regional application can limit model skill.

The current iteration of NEUSv2 meets these baseline calibration criteria for the simulated hindcast. Furthermore, the sensitivity analyses presented here indicate that the simulated functional groups are sensitive to changes in harvest, they can recover from mild to moderate disturbances, and their contemporary biomass is contingent on the ordering of historical fishing pressure. The species insensitive to changes in fishing are accounted for by data gaps in forcing or initial conditions. Future development of NEUSv2 seek to improve the dynamics and spatial scale of simulated fleets and further incorporation of environmental drivers on populations.

6.1 Including Figures

	Species			Relative	Target		Average		F
Code	Name	Stable	Reasonable	Change	Min	Max	Biomass	Catch	Rate
SCA	Sea scallop	F	T	0.71	0.70	21.00	7.28	0.22	0.03
ZL	Carnivorous zooplankton	F		0.20			3.74	<0.01	<0.01
BWH	Baleen whales	F		0.19			2.53	<0.01	<0.01
RWH	Right whales	F		0.17			1.04	<0.01	<0.01
COD	Atlantic cod	F	T	0.16	0.00	16.00	5.14	0.03	0.01
WPF	Windowpane flounder	F	T	0.10	0.00	0.98	0.43	0.01	0.02
TYL	All tilefish	F	F	0.09	0.00	0.15	0.22	0.01	0.03
TAU	Tautog	F	T	0.08	0.00	0.13	0.04	<0.01	0.02
SB	Seabirds	F		-0.07			<0.01	<0.01	<0.01
RED	Acadian redfish	F	T	-0.06	0.00	27.00	0.26	0.04	0.14
DC	Carrion	F					<0.01	<0.01	<0.01
MA	Macroalgae	F					<0.01	<0.01	<0.01
MB	Microphtybenthos	F					<0.01	<0.01	<0.01
SG	Seagrass	F					<0.01	<0.01	<0.01

Species				Relative	Target		Average		F
Code	Name	Stable	Reasonable	Change	Min	Max	Biomass	Catch	Rate
WTF	Witch flounder	T	T	0.05	0.01	0.77	0.62	0.01	0.01
PLA	American plaice	T	T	0.04	0.00	1.40	0.57	0.01	0.02
WSK	Winter skate	T	T	-0.04	0.00	21.00	1.48	0.12	0.08
BMS	Shallow macrozoobenthos	T	F	0.04	0.00	8.10	71.64	0.36	<0.01
SCU	Scup	T	T	-0.04	0.00	7.10	1.25	0.16	0.13
OPT	Ocean pout	T	T	0.04	0.00	1.00	0.5	<0.01	<0.01
DOG	Spiny dogfish	T	T	-0.04	0.00	73.00	1.83	0.16	0.09
TUN	Other tunas	T	F	0.03	0.00	0.01	7.04	<0.01	<0.01
HAD	Haddock	T	T	-0.03	0.00	62.00	1.66	0.2	0.12
DSH	Other demersal sharks	T	T	-0.03	0.00	11.00	4.84	<0.01	<0.01
SWH	Small toothed whales	T		-0.02			0.06	<0.01	<0.01
TWH	Toothed whales	T		-0.02			0.03	<0.01	<0.01
LOB	Lobster	T	T	-0.02	0.00	7.00	3.44	0.31	0.09
NSH	Northern shrimp other pandalids	T	T	0.02	0.01	11.00	0.3	<0.01	<0.01
WHK	White hake	T	T	0.02	0.39	5.60	1.32	0.03	0.02
BLF	Bluefish	T	T	-0.02	0.42	22.00	1.78	0.22	0.12
GOO	Goosefish	T	T	0.02	0.06	3.30	2.12	0.12	0.06
PIN	Pinnipeds	T		0.02			0.01	<0.01	<0.01
WOL	Wolffish	T		0.02			0.22	<0.01	<0.01
MAK	Atlantic mackerel	T	F	-0.01	0.00	5.50	7.77	0.16	0.02
ZM	Mesozooplankton	T		0.01			30.57	<0.01	<0.01
BSB	Black sea bass	T	F	-0.01	0.00	0.42	0.5	0.07	0.14
ZG	Gelatinous zooplankton	T		-0.01			1.65	<0.01	<0.01
MPF	Migratory mesopelagic fish	T	T	-0.01	0.00	0.57	0.02	<0.01	<0.01
BG	Benthic grazer	T	F	-0.01	0.00	0.00	60.59	<0.01	<0.01
OSH	Other shrimps	T	F	0.01	0.00	0.12	0.56	<0.01	<0.01
BIL	Billfish	T	F	-0.01	0.00	0.14	2.03	<0.01	<0.01
SAL	Atlantic salmon	T	F	0.01	0.00	0.00	0.06	<0.01	<0.01
POL	Pollock	T	T	-0.01	0.00	22.00	1.76	0.08	0.04
WIF	Winter flounder	T	T	0.01	0.00	1.60	1.48	0.02	0.01
BUT	Butterfish	T	T	-0.01	0.00	23.00	0.8	0.03	0.04
FDE	Shallow demersal fish	T	T	-0.01	0.02	5.50	1.13	0.01	0.01
YTF	Yellowtail flounder	T	T	-0.01	0.00	4.10	0.62	0.01	0.02
HAL	Atlantic halibut	T	F	-0.01	0.00	0.50	0.63	<0.01	<0.01

	Species			Relative	Target		Average		F
Code	Name	Stable	Reasonable	Change	Min	Max	Biomass	Catch	Rate
ANC	Anchovies	T	T	-0.01	0.00	75.00	4.39	<0.01	<0.01
STB	Striped bass	T	T	-0.01	0.21	36.00	1.77	0.01	0.01
HER	Atlantic herring	T	F	-0.01	35.00	1,500.00	15.89	0.61	0.04
BO	Meiobenthos	T		0.01			961.94	<0.01	<0.01
BPF	Other benthopelagic fish	T	T	-0.01	0.00	34.00	2.44	0.01	<0.01
SDF	Atlantic states demersal fish	T	F	-0.01	0.00	0.00	15.14	<0.01	<0.01
INV	Invasive vertebrate species	T		-0.01			0.01	<0.01	<0.01
BD	Deposit Feeder	T		-0.00			42.35	<0.01	<0.01
SUF	Summer flounder	T	T	0.00	0.09	10.00	4.08	0.12	0.03
SMO	Smooth dogfish	T	T	-0.00	0.05	3.00	1.38	0.01	0.01
FOU	Fourspot flounder	T	T	-0.00	0.01	0.74	0.11	<0.01	<0.01
POR	Porbeagle shark	T	T	-0.00	0.00	1.10	0.06	<0.01	<0.01
FLA	Other flatfish	T	T	-0.00	0.00	0.11	0.03	<0.01	<0.01
ISQ	Illex squid	T	T	-0.00	0.00	21.00	0.08	<0.01	<0.01
FDF	Miscellaneous demersal fish	T	F	0.00	11.00	160.00	2.69	0.04	0.02
DRM	Drums and croakers	T	T	0.00	0.00	67.00	4.07	0.03	0.01
OHK	Offshore hake	T	T	-0.00	0.00	0.67	0.09	<0.01	<0.01
LSQ	Loligo squid	T	T	-0.00	0.00	12.00	0.33	<0.01	<0.01
CLA	Atlantic surf clam	T	T	-0.00	19.00	220.00	57.27	0.17	<0.01
SHK	Silver hake	T	T	0.00	0.00	34.00	13.66	0.07	0.01
BC	Benthic Carnivore	T	F	-0.00	0.00	0.21	64.65	<0.01	<0.01
RHK	Red hake	T	T	-0.00	0.00	4.60	1.52	0.02	0.01
SSH	Sandbar shark	T	T	-0.00	0.00	2.50	0.17	<0.01	<0.01
LSK	Little skate	T	T	0.00	0.00	14.00	2.48	0.1	0.04
DR	Refractory detritus	T		0.00			33.77	<0.01	<0.01
BB	Sediment Bacteria	T		0.00			0.63	<0.01	<0.01
BLS	Blue shark	T		0.00			0.03	<0.01	<0.01
BFT	Atlantic bluefin tuna	T		-0.00			0.24	<0.01	<0.01
SK	Northeast skate complex	T	F	0.00	7.80	120.00	2.09	0.12	0.08
BFF	Other benthic filter feeder	T	F	-0.00	0.00	0.03	131.76	<0.01	<0.01
DL	Labile detritus	T		0.00			1.22	<0.01	<0.01
PB	Pelagic Bacteria	T		0.00			0.41	<0.01	<0.01
ZS	Microzooplankton	T		0.00			30.1	<0.01	<0.01
DF	Dinoflagellates	T		0.00			9.14	<0.01	<0.01

	Species			Relative	Target		Average		F
Code	Name	Stable	Reasonable	Change	Min	Max	Biomass	Catch	Rate
PS	Pico-phytoplankton	T		0.00			39.71	<0.01	<0.01
PL	Diatom	T		0.00			44.13	<0.01	<0.01
PSH	Other pelagic sharks	T	T	0.00	0.00	3.80	1.55	<0.01	<0.01
MEN	Atlantic menhaden	T	T	0.00	28.00	2,600.00	51.48	0.76	0.01
QHG	Ocean quahog	T	F	0.00	120.00	300.00	37.95	0.14	<0.01

Caracappa, J.C., Beet, A., Gaichas, S., Gamble, R.J., Hyde, K.J.W., Large, S.I., Morse, R.E., Stock, C.A., Saba, V.S., 2022. A northeast United States Atlantis marine ecosystem model with ocean reanalysis and ocean color forcing. Ecological Modelling 471, 110038. https://doi.org/10.1016/j.ecolmodel.2022.110038

Fay, G., DePiper, G., Steinback, S., Gamble, R.J., Link, J.S., 2019. Economic and Ecosystem Effects of Fishing on the Northeast US Shelf. Frontiers in Marine Science 6, 133. https://doi.org/10.3389/fmars.2019.00133

Fay, G., Link, J.S., Hare, J.A., 2017. Assessing the effects of ocean acidification in the Northeast US using an end-to-end marine ecosystem model. Ecological Modelling 347, 1–10. https://doi.org/10.1016/j.ecolmodel.2016.12.016

Fulton E. A., 2004. Ecological indicators of the ecosystem effects of fishing: Final report. CSIRO ; Australian Fisheries Management Authority, Hobart, Canberra.

Fulton, E.A., 2010. Approaches to end-to-end ecosystem models. Journal of Marine Systems 81, 171–183. https://doi.org/10.1016/j.jmarsys.2009.12.012

Fulton, E.A., Link, J.S., Kaplan, I.C., Savina-Rolland, M., Johnson, P., Ainsworth, C., Horne, P., Gorton, R., Gamble, R.J., Smith, A.D.M., Smith, D.C., 2011. Lessons in modelling and management of marine ecosystems: The Atlantis experience. Fish and Fisheries 12, 171–188. https://doi.org/10.1111/j.1467-2979.2011.00412.x

Jean-Michel, L., Eric, G., Romain, B.-B., Gilles, G., Angélique, M., Marie, D., Clément, B., Mathieu, H., Olivier, L.G., Charly, R., Tony, C., Charles-Emmanuel, T., Florent, G., Giovanni, R., Mounir, B., Yann, D., Pierre-Yves, L.T., 2021. The Copernicus Global 1/12° Oceanic and Sea Ice GLORYS12 Reanalysis. Frontiers in Earth Science 9, 698876. https://doi.org/10.3389/feart.2021.698876

Kaplan, I.C., Horne, P.J., Levin, P.S., 2012. Screening California Current fishery management scenarios using the Atlantis end-to-end ecosystem model. Progress in Oceanography 102, 5–18. https://doi.org/10.1016/j.pocean.2012.03.009

Kaplan, I.C., Marshall, K.N., 2016. A guinea pig's tale: Learning to review end-to-end marine ecosystem models for management applications. ICES Journal of Marine Science 73, 1715–1724. https://doi.org/10.1093/icesjms/fsw047

Link, J.S., Fulton, E.A., Gamble, R.J., 2010. The northeast US application of ATLANTIS: A full system model exploring marine ecosystem dynamics in a living marine resource management context. Progress in Oceanography 87, 214–234. https://doi.org/10.1016/j.pocean.2010.09.020

Liu, O.R., Kaplan, I.C., Hernvann, P.-Y., Fulton, E.A., Haltuch, M.A., Harvey, C.J., Marshall, K.N., Muhling, B., Norman, K., Pozo Buil, M., Rovellini, A., Samhouri, J.F., 2025. Climate Change Influences via Species Distribution Shifts and Century-Scale Warming in an End-To-End California Current Ecosystem Model. Global Change Biology 31, e70021. https://doi.org/10.1111/gcb.70021

Olsen, E., Fay, G., Gaichas, S., Gamble, R., Lucey, S., Link, J.S., 2016. Ecosystem Model Skill Assessment. Yes We Can! PLOS ONE 11, e0146467. https://doi.org/10.1371/journal.pone.0146467

Rose, K.A., Allen, J.I., Artioli, Y., Barange, M., Blackford, J., Carlotti, F., Cropp, R., Daewel, U., Edwards, K., Flynn, K., Hill, S.L., HilleRisLambers, R., Huse, G., Mackinson, S., Megrey, B., Moll, A., Rivkin, R., Salihoglu, B., Schrum, C., Shannon, L., Shin, Y.-J., Smith, S.L., Smith, C., Solidoro, C., St. John, M., Zhou, M., 2010. End-To-End Models for the Analysis of Marine Ecosystems: Challenges, Issues, and Next Steps. Marine and Coastal Fisheries 2, 115–130. https://doi.org/10.1577/C09-059.1

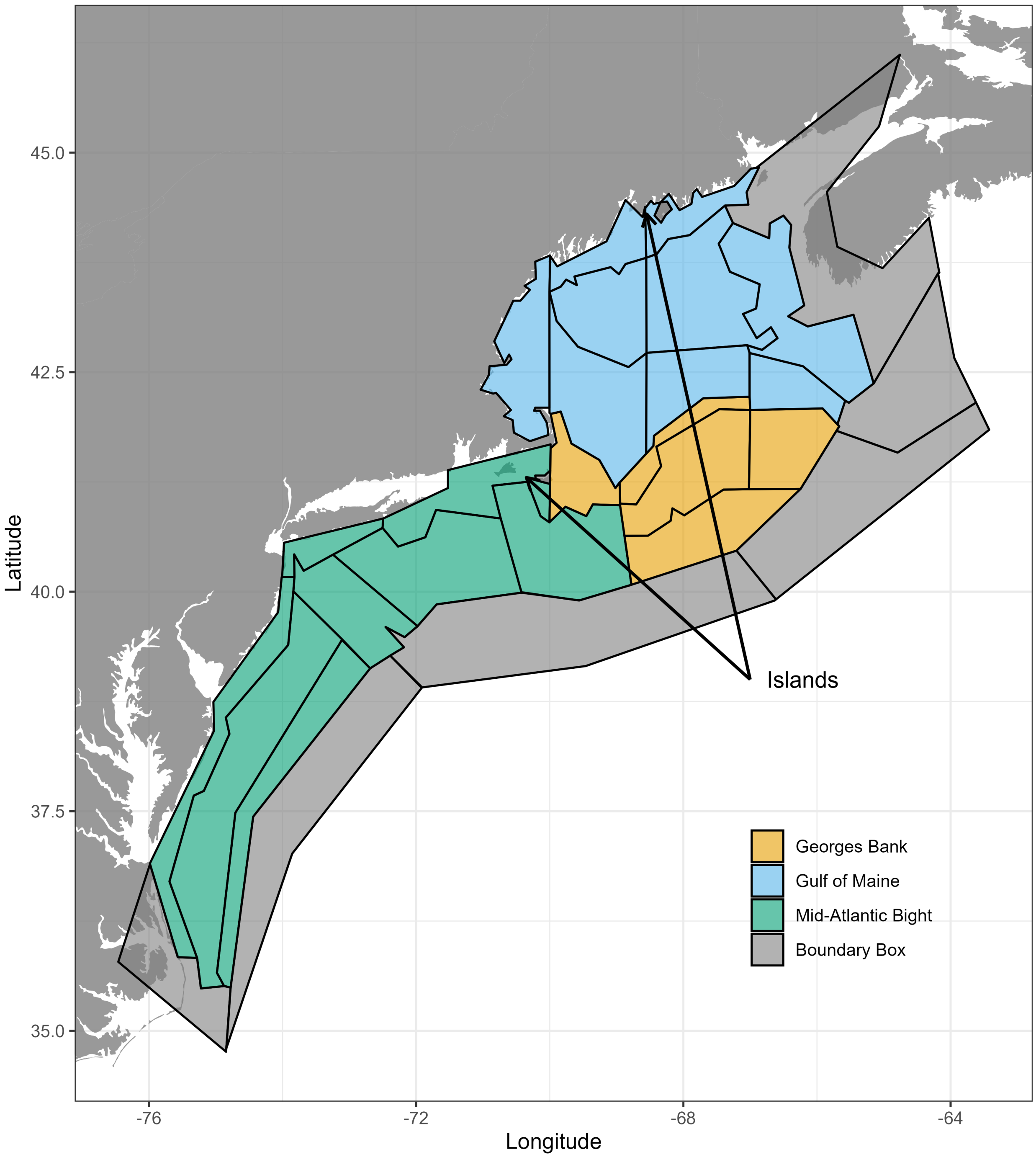

Figure 1: NEUSv2 domain and regional subdivisions (colors).

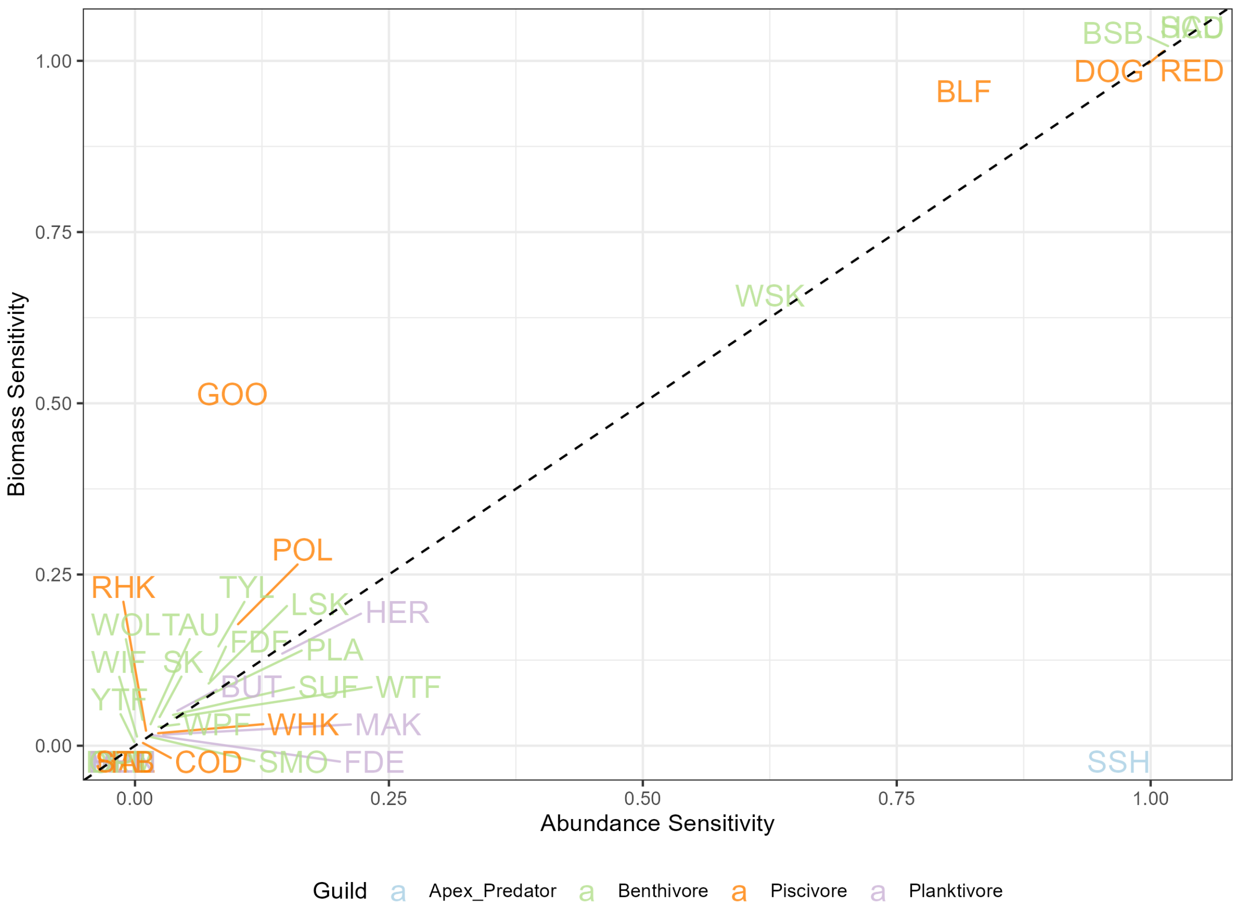

Figure 2: Comparisons of sensitivity metrics for biomass ($R_{biomass}$) and abundance ($R_{abundance}$) as defined in eq ef:responsiveness: Individual species are marked by their shorthand code and guild assignment (colors).

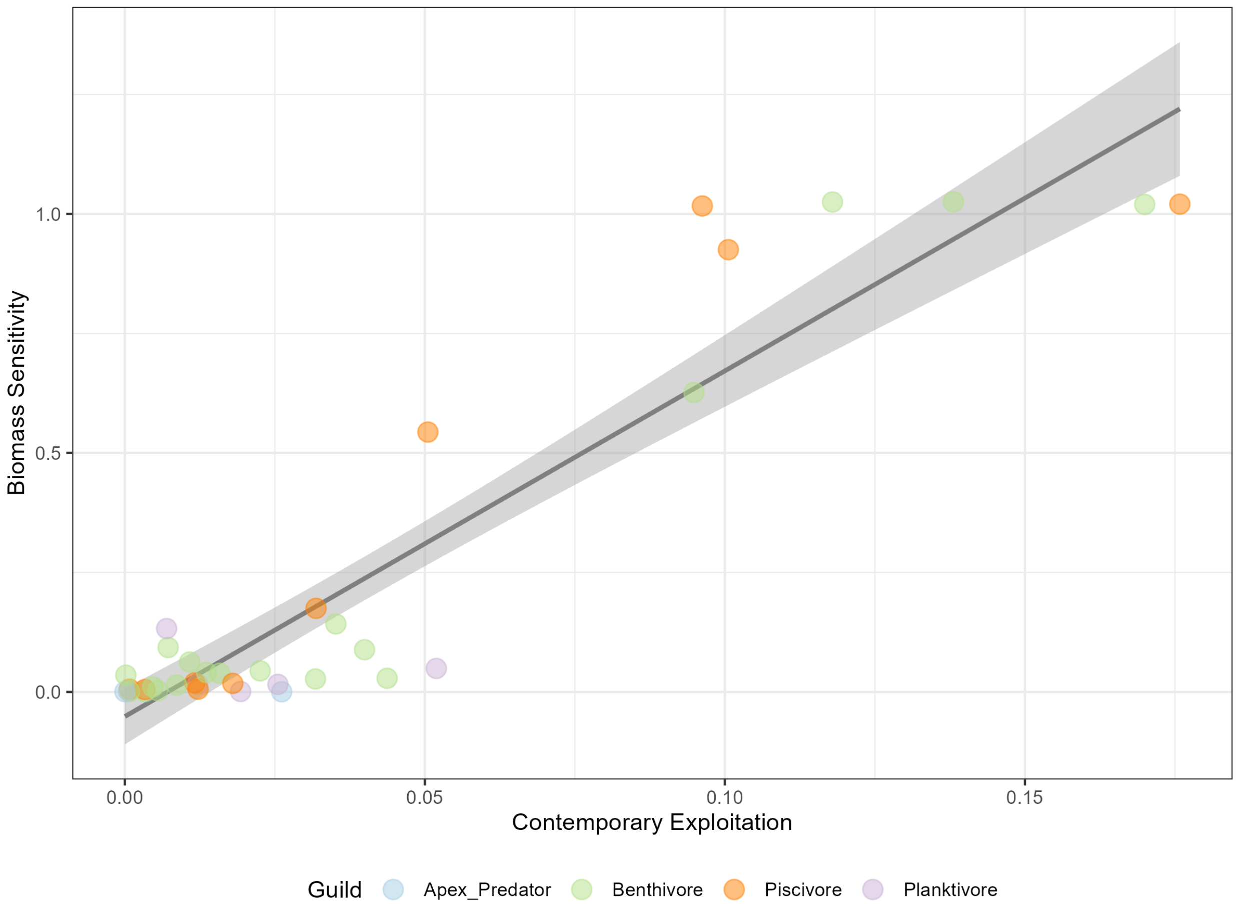

Figure 3: Regression of contemporary exploitation rate in the baseline simulation and biomass sensitivity. Individual species are marked (points) as well as their guild (colors) with a linear regression (black line) and 95% confidence interval (grey area).

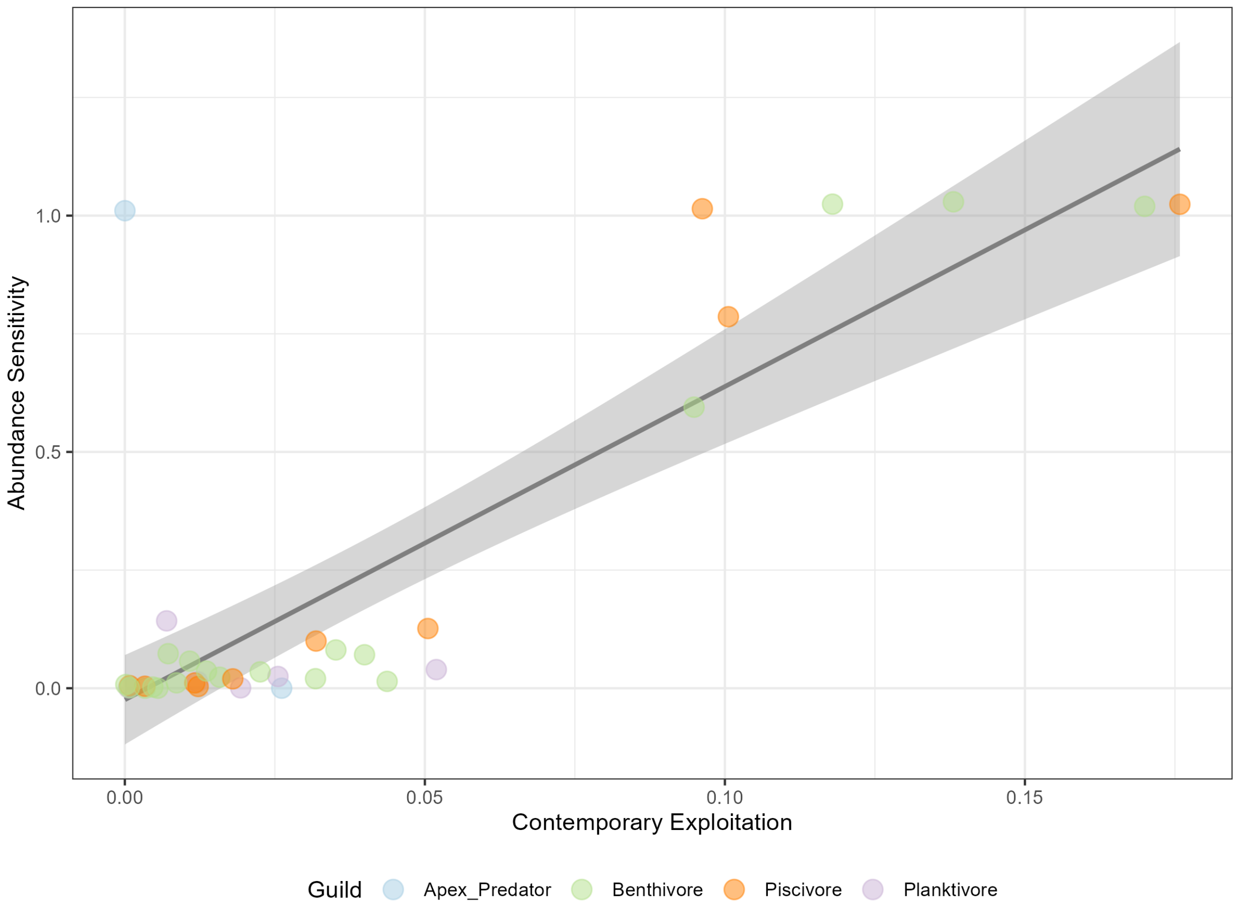

Figure 4: Regression of contemporary exploitation rate in the baseline simulation and abundance sensitivity. Individual species are marked (points) as well as their guild (colors) with a linear regression (black line) and 95% confidence interval (grey area).

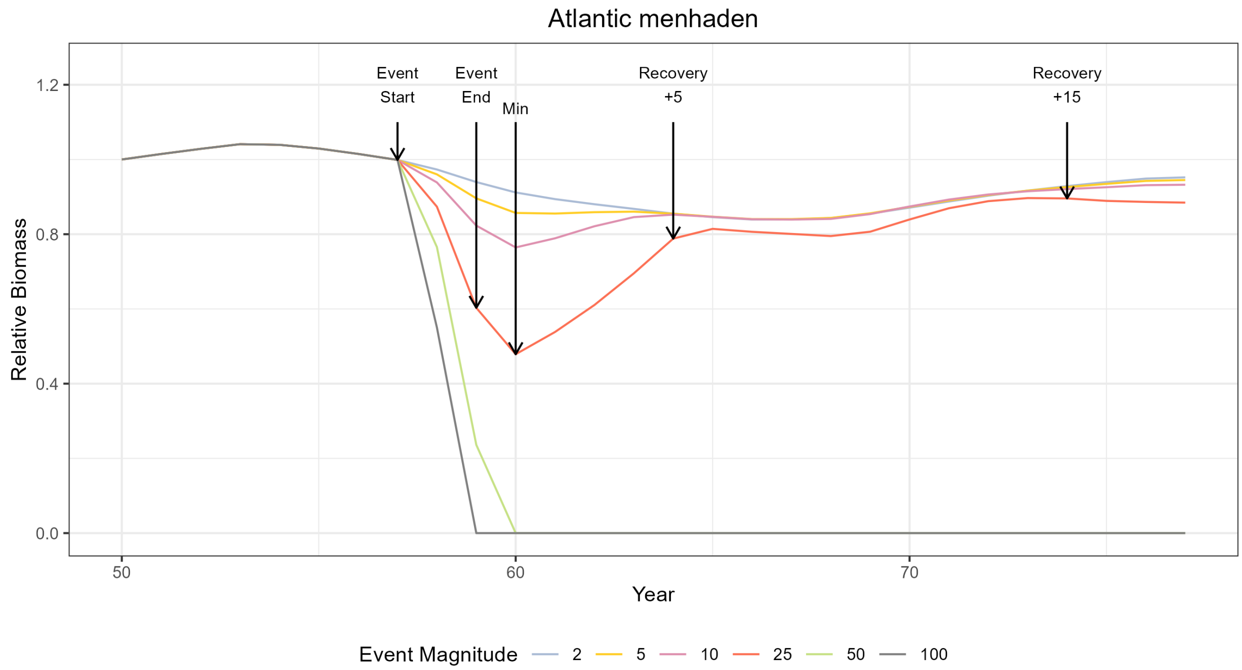

Figure 5: An example schematic of the disturbance-recovery scenario definitions. All scenarios have the same output through year 57. At “event start” different fishing forcing is applied at different magnitudes (colored lines). The “event end” occurs after 2 years. Then recovery is assessed 5 and 15 years after the event end.

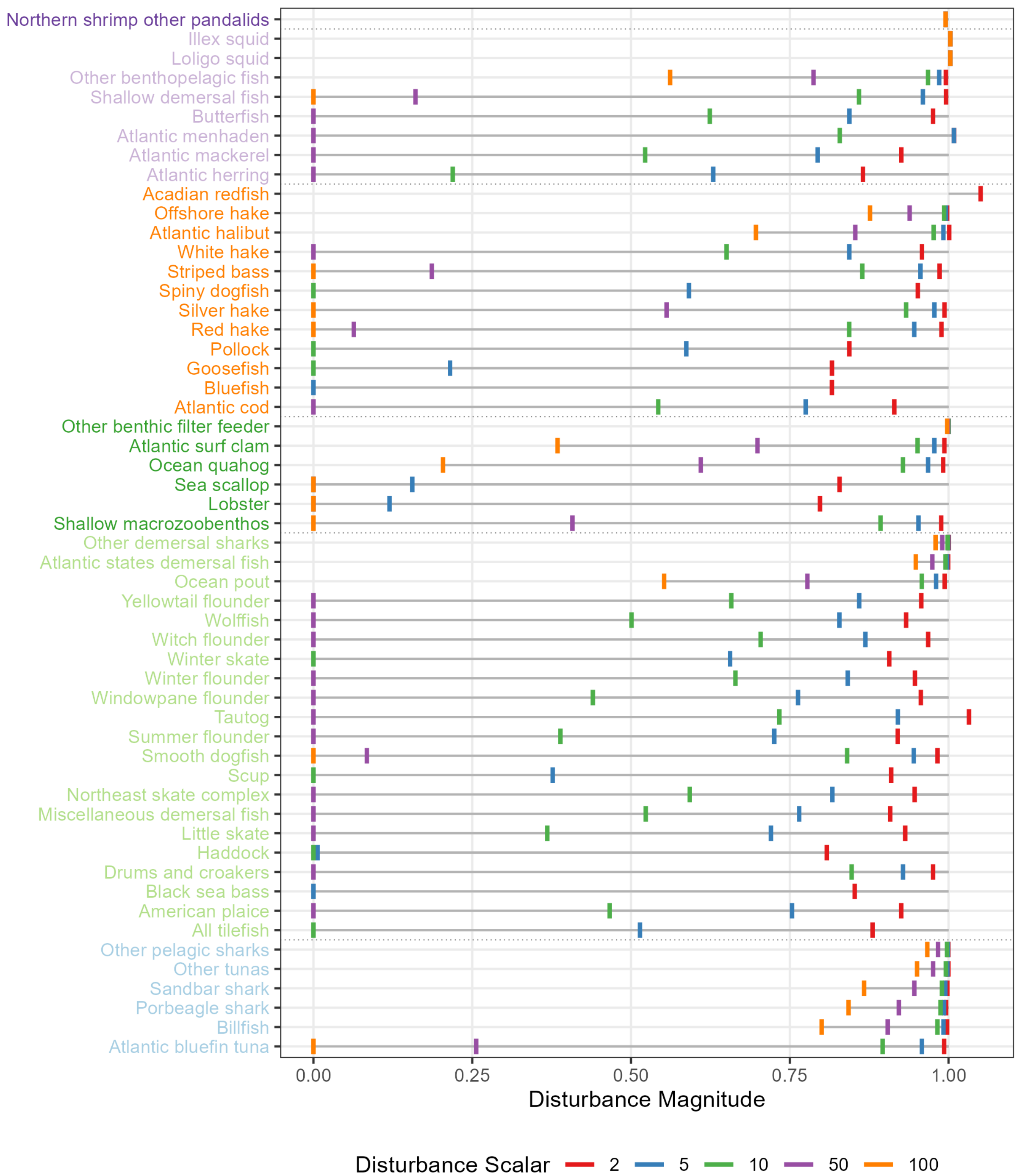

Figure 6: Measurements of initial disturbance magnitude (vertical lines) relative to the event start time at various disturbance scalars (colors). Species are grouped according to their guild (text color).

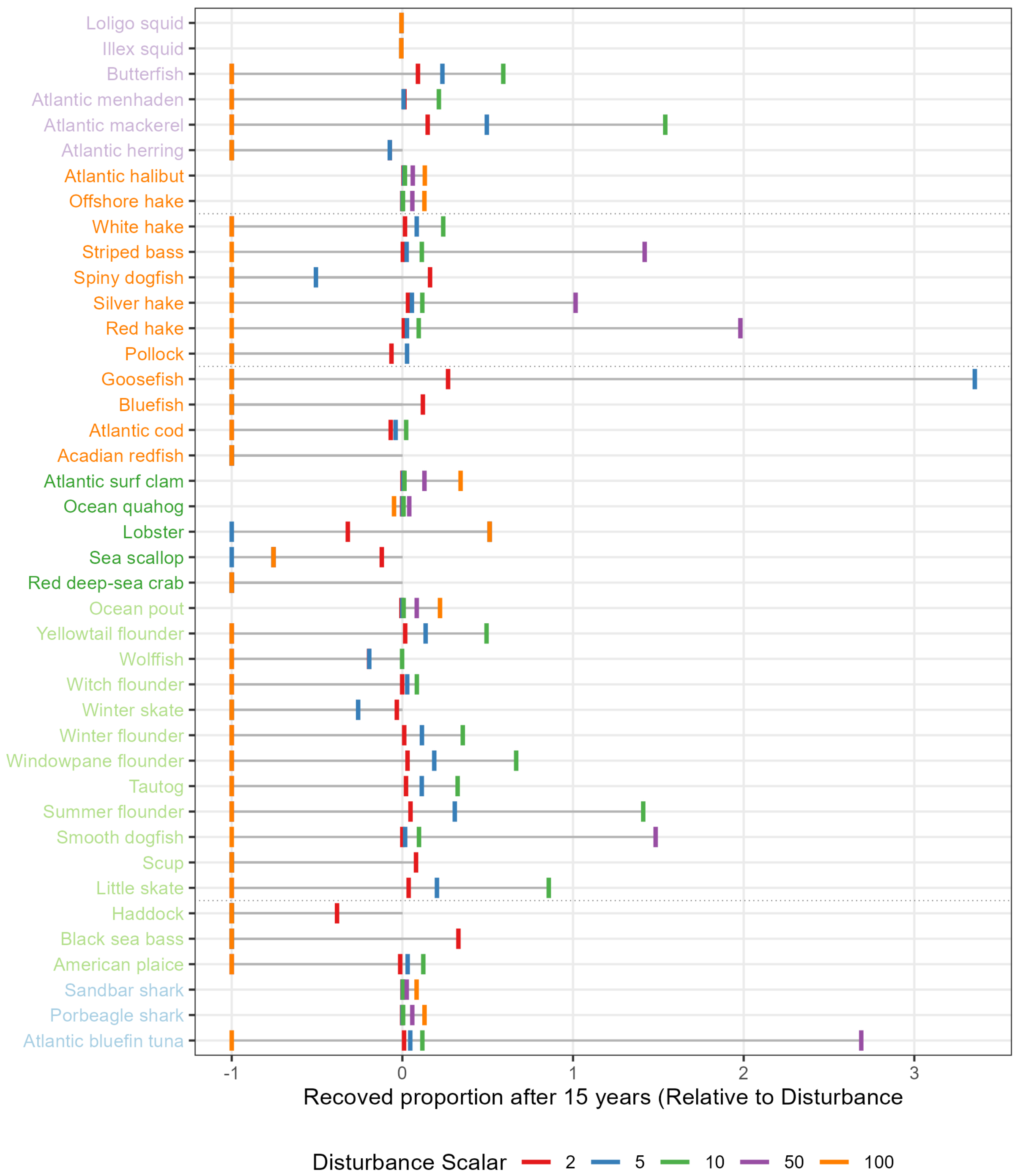

Figure 7: Recovered proportion of event start biomass (vertical lines) for different disturbance scalars (bar colors). Species are grouped by guild (text colors).

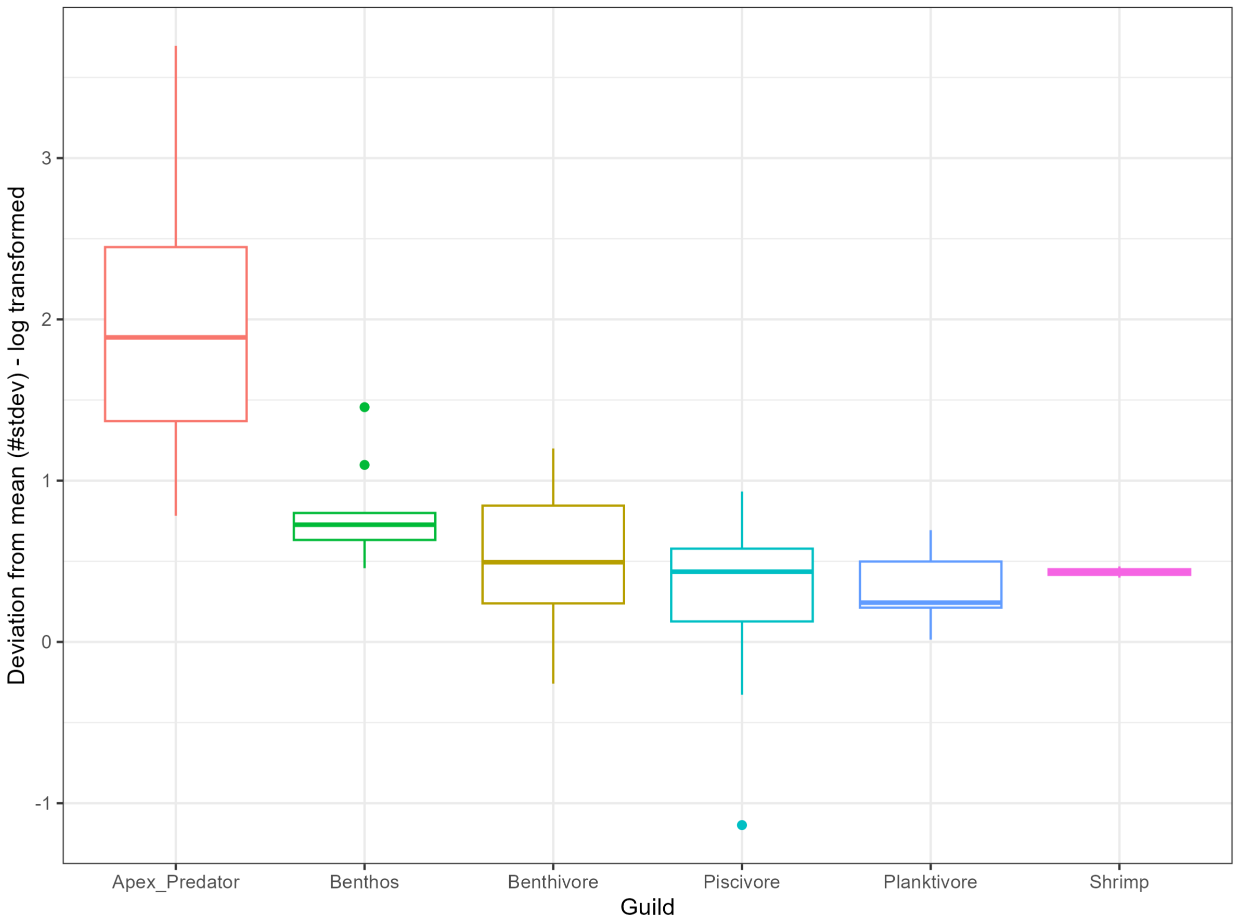

Figure 8: Box plots show the deviation of biomass in the baseline run from the distribution of randomized catch runs in terms of standard deviations from the mean across species within a guild. A deviation of zero indicates that the mean biomass of the randomized runs is identical to the baseline run.

Rovellini, A., Punt, A.E., Bryan, M.D., Kaplan, I.C., Dorn, M.W., Aydin, K., Fulton, E.A., Alglave, B., Baker, M.R., Carroll, G., Ferriss, B.E., Haltuch, M.A., Hayes, A.L., Hermann, A.J., Hernvann, P.-Y., Holsman, K.K., Liu, O.R., McHuron, E., Morzaria-Luna, H.N., Moss, J., Surma, S., Weise, M.T., 2025. Linking climate stressors to ecological processes in ecosystem models, with a case study from the Gulf of Alaska. ICES Journal of Marine Science 82, fsae002. https://doi.org/10.1093/icesjms/fsae002

Turner, K.J., Mouw, C.B., Hyde, K.J.W., Morse, R., Ciochetto, A.B., 2021. Optimization and assessment of phytoplankton size class algorithms for ocean color data on the Northeast U.S. Continental shelf. Remote Sensing of Environment 267, 112729. https://doi.org/10.1016/j.rse.2021.112729